

Deriving Nonlinear Correlation Coefficients from Partial Moments

By

Fred Viole
OVVO Financial Systems
fred.viole@ovvofinancialsystems.com

And

David Nawrocki
Villanova University
Villanova School of Business
800 Lancaster Avenue
Villanova, PA 19085 USA
610-519-4323
david.nawrocki@villanova.edu

Deriving Nonlinear Correlation Coefficients from Partial Moments

Abstract

We introduce a nonlinear correlation coefficient metric derived from partial moments that can be substituted for the Pearson correlation coefficient in linear instances as well. The flexibility offered by partial moments enables ordered partitions of the data whereby linear segments are aggregated for an overall correlation coefficient. Our coefficient works without the need to perform a linear transformation on the underlying data, and can also provide a general measure of nonlinearity between two variables. We also extend the analysis to a multiple nonlinear regression without the adverse effects of multicollinearity.

Keywords: Partial moments, Nonlinear correlation, Nonlinear multivariate analysis

1. INTRODUCTION

Chen et al. (2010) explore the problem of estimating a nonlinear correlation (See Figure 1). They note that a generic use statistic such as the Pearson correlation coefficient does not exist for nonlinear correlations. We introduce a generic nonlinear correlation coefficient metric derived from partial moments that can be substituted for the Pearson correlation coefficient in linear instances as well. The flexibility offered by partial moments enables ordered partitions of the data whereby linear segments are aggregated for an overall correlation coefficient.

Partial moments have three main advantages: (1) no distributional assumption is required, (2) partial moments are integrated into economics through expected utility theory (Holthausen, 1981 and Guthoff et al., 1997), and are integrated into statistics as Viole and Nawrocki (2012a) find that partial moments can be used to derive the CDF and PDF of any distribution.

The paper is organized as follows: The next section will cover the development of the measure followed by a section with empirical results. Next, we extend the analysis to a multidimensional nonlinear analysis with an application to nonlinear regression analysis. A final discussion and summary completes the paper.

2. DEVELOPMENT OF NONLINEAR CORRELATION MEASURE

The Pearson correlation coefficient is represented by

$$\rho_{x,y} = \frac{cov(X,Y)}{\sigma_x \sigma_y}$$

and is standardized in the range [-1,1]. The covariance and standard deviation cannot isolate and differentiate the information present in each of the four possible relationships between two variables where the target is some reference point:

X below target, *Y* below target

X below target, *Y* above target

X above target, *Y* below target

X above target, *Y* above target

We propose a method of partitioning the distribution with partial moments to capture the information from each linear relationship embedded within a bi- or multivariate relationship (linear or nonlinear). Based on the above four relationships between two variables, a co- or divergent partial moment is constructed to quantify it.ⁱ

2.1 Co-Partial Moments

$$CLPM(n, h_x | h_y, X | Y) = \frac{1}{T} \left[\sum_{t=1}^T (\max\{0, h_x - X_t\}^n \cdot \max\{0, h_y - Y_t\}^n) \right] \quad (1)$$

$$CUPM(q, l_x | l_y, X | Y) = \frac{1}{T} \left[\sum_{t=1}^T (\max\{X_t - l_x, 0\}^q \cdot \max\{0, Y_t - l_y\}^q) \right] \quad (2)$$

where X_t represents the observation X at time t , Y_t represents the observation Y at time t , n is the degree of the LPM, q is the degree of the UPM, h_x is the target for computing below target observations for X , and l_x is the target for computing above target observations for X . For simplicity we assume that $h_x = l_x$.

2.2 Divergent Partial Moments

$$DLPM(q | n, h_x | h_y, X | Y) = \frac{1}{T} \left[\sum_{t=1}^T (\max\{X_t - h_x, 0\}^q \cdot \max\{0, h_y - Y_t\}^n) \right] \quad (3)$$

$$DUPM(n | q, h_x | h_y, X | Y) = \frac{1}{T} \left[\sum_{t=1}^T (\max\{0, h_x - X_t\}^n \cdot \max\{Y_t - h_y, 0\}^q) \right] \quad (4)$$

2.3 Definition of Variable Relationships:

X below target, Y below target $\rightarrow CLPM(n, h_x|h_y, X|Y)$

X below target, Y above target $\rightarrow DUPM(n|q, h_x|h_y, X|Y)$

X above target, Y below target $\rightarrow DLPM(q|n, h_x|h_y, X|Y)$

X above target, Y above target $\rightarrow CUPM(q, h_x|h_y, X|Y)$

To avoid the blunt covariance and standard deviation dependence of the Pearson coefficient, we can use the following nonparametric formula in equation 5 to determine the correlation (linear or nonlinear) between two variables.

$$\rho_{xy} = \frac{[CLPM(0, h_x|h_y, x|y) - DLPM(0, h_x|h_y, x|y) - DUPM(0, h_x|h_y, x|y) + CUPM(0, h_x|h_y, x|y)]}{[CLPM(0, h_x|h_y, x|y) + DLPM(0, h_x|h_y, x|y) + DUPM(0, h_x|h_y, x|y) + CUPM(0, h_x|h_y, x|y)]} \quad (5)$$

The axiomatic relationship between correlation and co- or divergent returns follows. If there is a -1 correlation, then the returns between the variables will always be divergent, thus,

$$\rho_{xy} = \frac{[0 - DLPM(0, h_x|h_y, x|y) - DUPM(0, h_x|h_y, x|y) + 0]}{[0 + DLPM(0, h_x|h_y, x|y) + DUPM(0, h_x|h_y, x|y) + 0]} = -1 \quad (6)$$

If there is a perfect correlation between two variables, then there will be no divergent returns, thus,

$$\rho_{xy} = \frac{[CLPM(0, h_x|h_y, x|y) - 0 - 0 + CUPM(0, h_x|h_y, x|y)]}{[CLPM(0, h_x|h_y, x|y) + 0 + 0 + CUPM(0, h_x|h_y, x|y)]} = 1 \quad (7)$$

If there is zero correlation between two variables, then the co- and divergent returns will be of equal frequency or magnitude (degree zero and degree one respectively),

$$CLPM(0, h_x|h_y, x|y) = DLPM(0, h_x|h_y, x|y) = DUPM(0, h_x|h_y, x|y) = CUPM(0, h_x|h_y, x|y)$$

Thus,

$$\begin{aligned} \rho_{xy} &= \\ &\frac{[CLPM(0, h_x|h_y, x|y) - DLPM(0, h_x|h_y, x|y) - DUPM(0, h_x|h_y, x|y) + CUPM(0, h_x|h_y, x|y)]}{[CLPM(0, h_x|h_y, x|y) + DLPM(0, h_x|h_y, x|y) + DUPM(0, h_x|h_y, x|y) + CUPM(0, h_x|h_y, x|y)]} \\ &= 0 \end{aligned} \quad (8)$$

Degree one can be substituted for parameters n and q , to generate correlations whereby the magnitude of the target deviations are compared; thus generating a dependence coefficient.

2.4 Visualization of the Partitions Using Means as Targets:

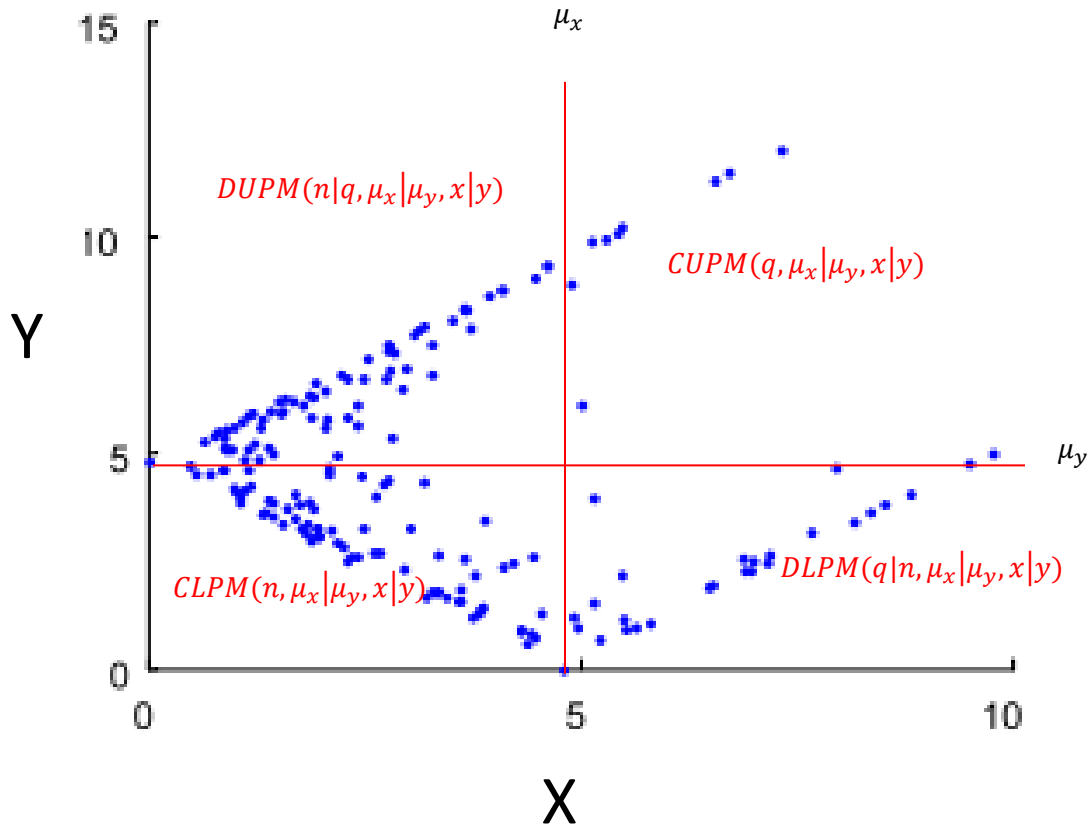


Figure 1. 1st order partitioning of the distribution based on variable relationships with co- and divergent partial moments on an observed nonlinear correlation in a microarray study from Chen et al. (2010).

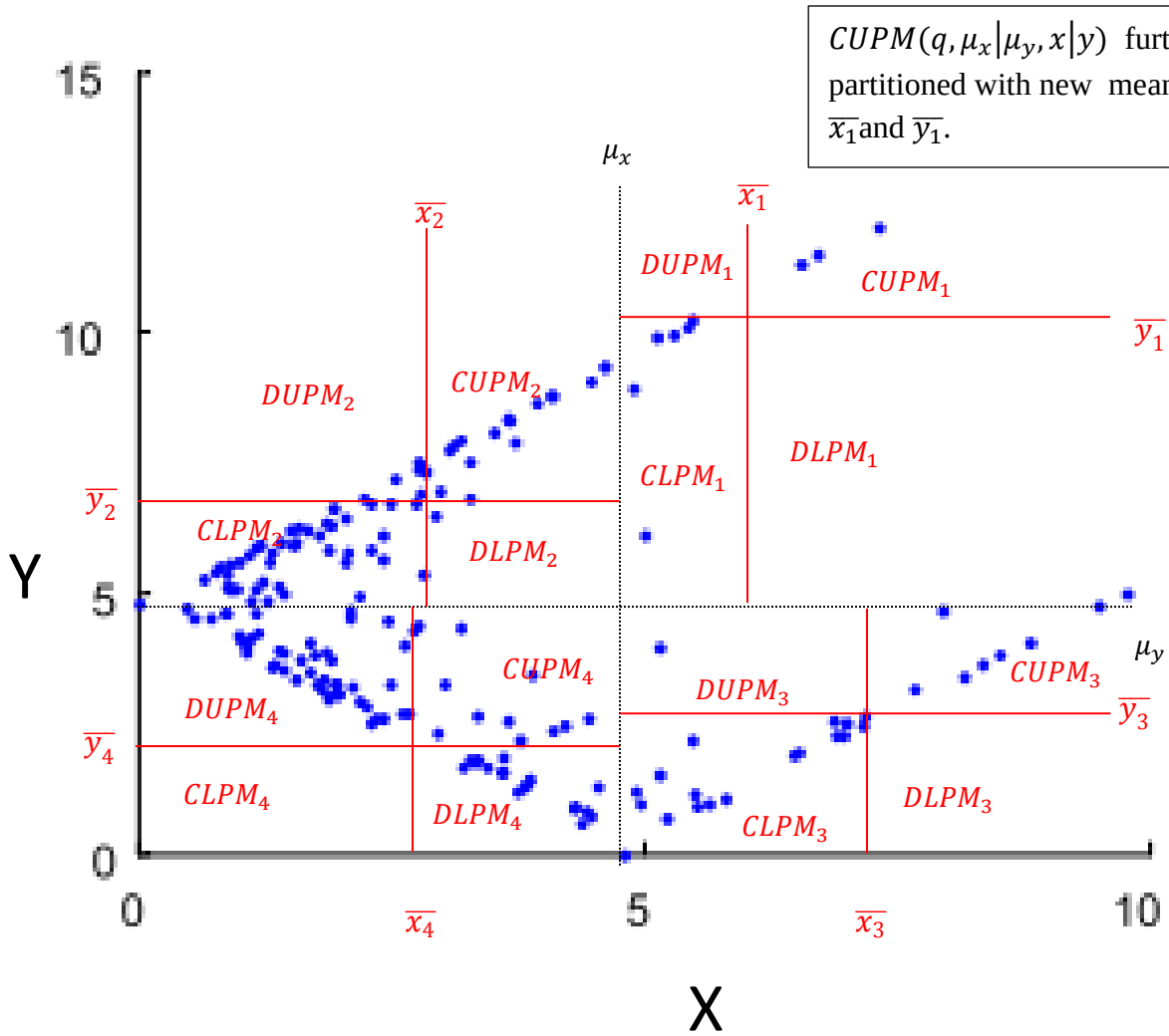


Figure 2. 2nd order partitioning of the microarray study based on means of partial moment subsets as targets.

2.5 Definition of Variable Subsets:

$$\{x_1, y_1\} \in CUPM(q, \mu_x | \mu_y, x | y)$$

$$\{x_2, y_2\} \in DLPM(q | n, \mu_x | \mu_y, x | y)$$

$$\{x_3, y_3\} \in CLPM(n, \mu_x | \mu_y, x | y)$$

$$\{x_4, y_4\} \in DUPM(n | q, \mu_x | \mu_y, x | y)$$

2.6 Definition of Subset Means:

$$\begin{aligned}\bar{x}_1 &= \frac{\sum_{n=1}^n x_{1n}}{n} & \bar{y}_1 &= \frac{\sum_{n=1}^n y_{1n}}{n} \\ \bar{x}_2 &= \frac{\sum_{n=1}^n x_{2n}}{n} & \bar{y}_2 &= \frac{\sum_{n=1}^n y_{2n}}{n} \\ \bar{x}_3 &= \frac{\sum_{n=1}^n x_{3n}}{n} & \bar{y}_3 &= \frac{\sum_{n=1}^n y_{3n}}{n} \\ \bar{x}_4 &= \frac{\sum_{n=1}^n x_{4n}}{n} & \bar{y}_4 &= \frac{\sum_{n=1}^n y_{4n}}{n}\end{aligned}$$

2.7 Defintion of Subset Partial Moments:

$$CUPM_1(q, \bar{x}_1 | \bar{y}_1, X_1 | Y_1) = \frac{1}{T} \left[\sum_{t=1}^T (\max\{0, x_{1t} - \bar{x}_1\}^q \cdot \max\{0, y_{1t} - \bar{y}_1\}^q) \right] \quad (9)$$

$$DLPM_1(q | n, \bar{x}_1 | \bar{y}_1, X_1 | Y_1) = \frac{1}{T} \left[\sum_{t=1}^T (\max\{x_{1t} - \bar{x}_1, 0\}^q \cdot \max\{0, \bar{y}_1 - y_{1t}\}^n) \right] \quad (10)$$

$$CLPM_1(n, \bar{x}_1 | \bar{x}_1, X_1 | Y_1) = \frac{1}{T} \left[\sum_{t=1}^T (\max\{0, \bar{x}_1 - x_{1t}\}^n \cdot \max\{0, \bar{y}_1 - y_{1t}\}^n) \right] \quad (11)$$

$$DUPM_1(n | q, \bar{x}_1 | \bar{y}_1, X_1 | Y_1) = \frac{1}{T} \left[\sum_{t=1}^T (\max\{0, \bar{x}_1 - x_{1t}\}^n \cdot \max\{y_{1t} - \bar{y}_1, 0\}^q) \right] \quad (12)$$

For a 3rd order analysis for example, one needs to then compute the 12 remaining subset partial moments (in addition to the four identified in equations 9-12 above) using the appropriate subset mean targets for each quadrant. The total amount of subset means will be less than or equal to $4^{(N-1)}$ where N is the number of orders specified.ⁱⁱ

The eventual correlation metric is accomplished by adding all CUPM's and CLPM's (positive correlations) and subtracting DUPM's and DLPM's (negative correlations) in the numerator, while summing all 16 co- and divergent partial moments representing the entire distribution in the denominator per equation 13 below.

$$\rho_{xy} =$$

Numerator:

$$(CLPM_1 + CLPM_2 + CLPM_3 + CLPM_4 - DLPM_1 - DLPM_2 - DLPM_3 - DLPM_4 - DUPM_1 - DUPM_2 - DUPM_3 - DUPM_4 + CUPM_1 + CUPM_1 + CUPM_3 + CUPM_4)$$

Denominator:

$$(CLPM_1 + CLPM_2 + CLPM_3 + CLPM_4 + DLPM_1 + DLPM_2 + DLPM_3 + DLPM_4 + DUPM_1 + DUPM_2 + DUPM_3 + DUPM_4 + CUPM_1 + CUPM_1 + CUPM_3 + CUPM_4)$$

(13)

2.8 Dependence:

We can also define the dependence present between two variables as the sum of the absolute value of the per quadrant correlations. Stated differently, when all of the per quadrant observations are either the CLPM & CLPM, or DLPM & DUPM, the variables are dependent upon one another.

$$\eta(X,Y) = |\rho_{CLPM}| + |\rho_{CUPM}| + |\rho_{DLPM}| + |\rho_{DUPM}| \quad (14)$$

Where the CLPM quadrant's correlation is given by

$$|\rho_{CLPM}| = \left| \frac{CLPM_4 + CUPM_4 - DLPM_4 - DUPM_4}{CLPM_4 + CUPM_4 + DLPM_4 + DUPM_4} \right|$$

Equation 14 describes the amount of nonlinearity present in each quadrant when the negative correlations are equal in frequency or magnitude (depending on degree 0 or 1 respectively) to the positive correlations.

When $\eta(X,Y)$ equals one, there is maximum *dependence* between the two variables. As $\eta(X,Y)$ approaches 0, the relationship is approaching maximum *independence*.

3. EMPIRICAL EVIDENCE:

Third order partitions are shown and calculated in R. The 1st order partition is the thick red line (per Figure 1), the 2nd order partition is the thin red line (per Figure 2) and the 3rd order partition is the dotted black line.

Linear Equalities:ⁱⁱⁱ

$$Y = 2X$$

```
> x=seq(-3,3,.01);y=2*x
> cor(x,y)
[1] 1
> NNS.dep(x,y,print.map = T)
$Correlation
[1] 1
$Dependence
[1] 1
```

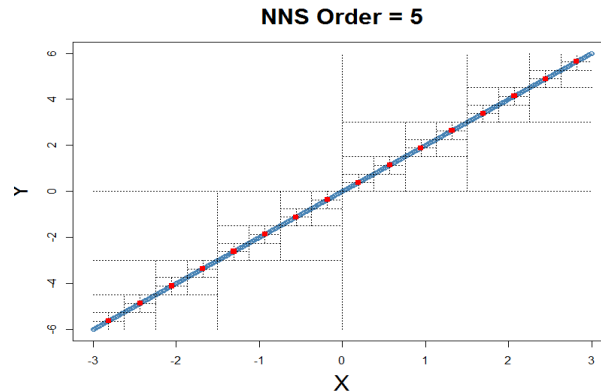


Figure 3. Linear positive relationship between two variables (X, Y).

$$Y = -2X$$

```
> x=seq(-3,3,.01);y=-2*x
> cor(x,y)
[1] -1
> NNS.dep(x,y,print.map = T)
$Correlation
[1] -1
$Dependence
[1] 1
```

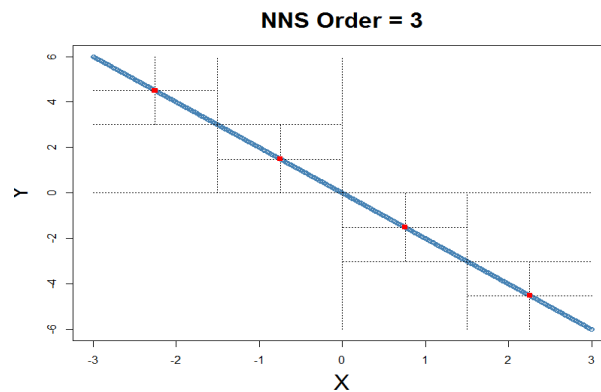


Figure 4. Linear inverse relationship between two variables (X, Y).

Nonlinear Differences:

$Y = X^2$ for positive X

```
> x=seq(0,3,.01);y=x^2
> cor(x,y)
[1] 0.9680452
> NNS.dep(x,y,print.map = T)
$Correlation
[1] 0.9994402
$Dependence
[1] 0.9994402
```

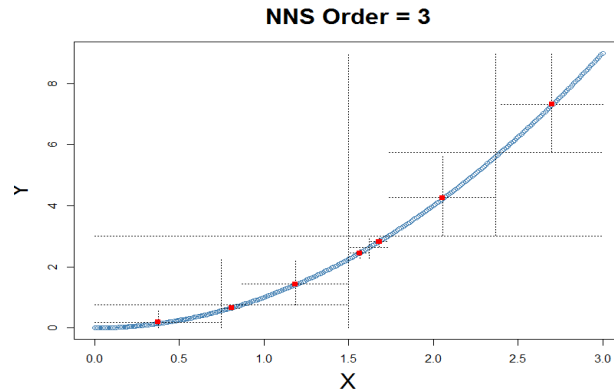


Figure 5. Nonlinear positive relationship between two variables (X, Y).

$Y = X^2$

```
> x=seq(-3,3,.01);y=x^2
> cor(x,y)
[1] 7.665343e-17
> NNS.dep(x,y,print.map = T)
$Correlation
[1] -0.001647721
$Dependence
[1] 0.9993975
```

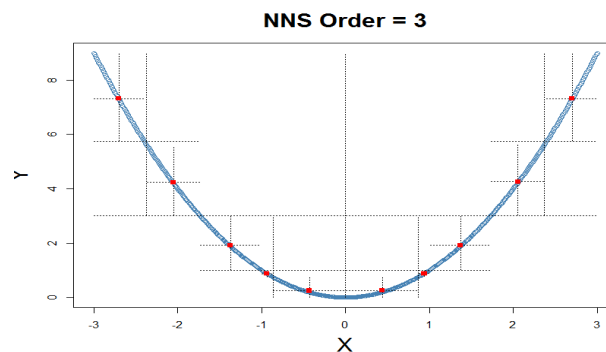


Figure 6. Nonlinear relationship between two variables (X, Y).

As the exponential function increases in magnitude, we actually find it to retain its linear relationship...

$$Y = X^{10}$$

```
> x=seq(0,3,.01);y=x^10
> cor(x,y)
[1] 0.6610183
> NNS.dep(x,y,print.map = T)
$Correlation
[1] 0.9812511
$Dependence
[1] 0.9812511
```

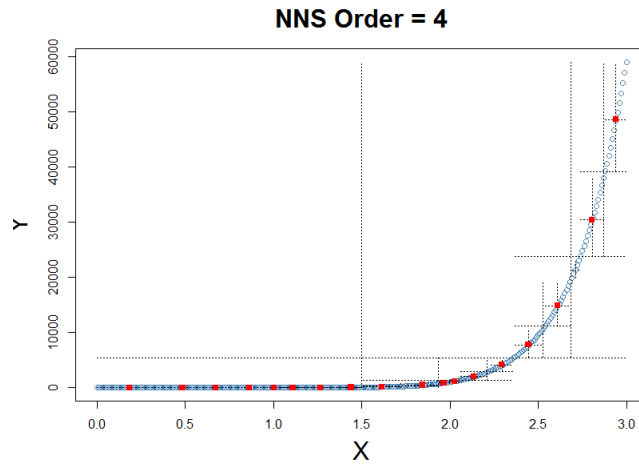


Figure 7. Nonlinear positive relationship between two variables (X, Y).

And a completely nonlinear clustered dataset, where coefficient weighting due to partition occupancy is exemplified.

$$Y = \text{undetermined } f(x)$$

```
>
cor(cluster.df[,3],cluster.df[,4])
[1] -0.6275592
> NNS.dep(cluster.df[,3],
cluster.df[,4],print.map = T)
$Correlation
[1] -0.1020994
$Dependence
[1] 0.2637387
```

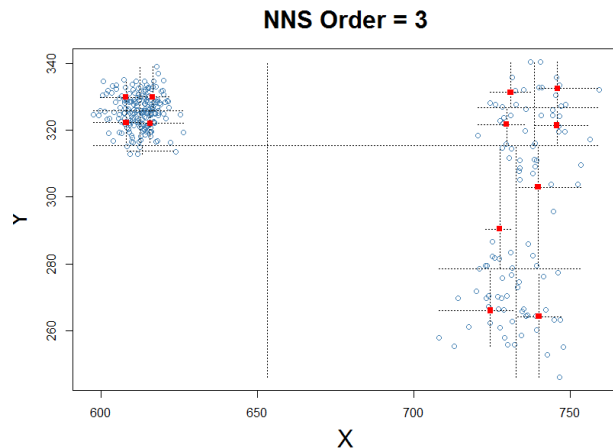


Figure 8. Nonlinear relationship between two variables (X, Y).

4. MULTIDIMENSIONAL NONLINEAR ANALYSIS:

To find the 1st order aggregate correlation for more than two dimensions, the method is similar to what was just presented. Instead of co- and divergent partial moments, we are going to substitute co- and divergent partial moment *matrices* into equation 5. A $n \times n$ matrix for each of the interactions (CLPM, DLPM, DUPM and CUPM) per Viole and Nawrocki (2012a), can be constructed and treated analogously to the direct partial moment computation.

Thus,

$$CLPM_{matrix}(0, h_x \dots h_n, x \dots n) = \begin{pmatrix} CLPM(0, h_x | h_x, x | x) & \dots & CLPM(0, h_x | h_n, x | n) \\ \vdots & \ddots & \vdots \\ CLPM(0, h_n | h_x, n | x) & \dots & CLPM(0, h_n | h_n, n | n) \end{pmatrix} \quad (15)$$

Yielding,

$$\rho_{x \dots n} = \frac{[CLPM_{matrix}(0, h_x \dots h_n, x \dots n) - DLPM_{matrix}(0, h_x \dots h_n, x \dots n) - DUPM_{matrix}(0, h_x \dots h_n, x \dots n) + CUPM_{matrix}(0, h_x \dots h_n, x \dots n)]}{[CLPM_{matrix}(0, h_x \dots h_n, x \dots n) + DLPM_{matrix}(0, h_x \dots h_n, x \dots n) + DUPM_{matrix}(0, h_x \dots h_n, x \dots n) + CUPM_{matrix}(0, h_x \dots h_n, x \dots n)]} \quad (16)$$

Whereby the final result will be an equal sized $n \times n$ matrix,

$$\rho_{x \dots n} = \begin{pmatrix} \rho_{xx} & \dots & \rho_{xn} \\ \vdots & \ddots & \vdots \\ \rho_{nx} & \dots & \rho_{nn} \end{pmatrix} = \begin{pmatrix} 1 & \dots & \rho_{xn} \\ \vdots & \ddots & \vdots \\ \rho_{nx} & \dots & 1 \end{pmatrix}$$

To derive the overall correlation, we need to sterilize the main diagonal of 1's (which are self-correlations) with the following formula,

$$\rho_{x...n} = \frac{\left[\sum \begin{pmatrix} \rho_{xx} & \cdots & \rho_{xn} \\ \vdots & \ddots & \vdots \\ \rho_{nx} & \cdots & \rho_{nn} \end{pmatrix} - n \right]}{n^2 - n} \quad (17)$$

Again, if the variables are all below or above their respective targets at time t , the CLPM and CUPM matrices respectively will capture that information. If the variables are i.i.d., the likelihood that one variable would diverge at time t increases as n increases, reducing $\rho_{x...n}$.

Further order partition analysis can be translated to the multidimensional by creating matrices for each of the identified subsets for all of the variables.

4.1 Nonlinear Regression Analysis:

The target means from which the four partial moment matrices are calculated also serve as the basis for a nonlinear regression. By plotting all of the mean intersections, the linear segments will fit the underlying, nonparametrically. The increased order of portioning will generate more intersecting points (maximum of $4^{(N-1)}$) for a more granular analysis. Below is an example with 3rd order partitioning, generating a fit to the linear data.

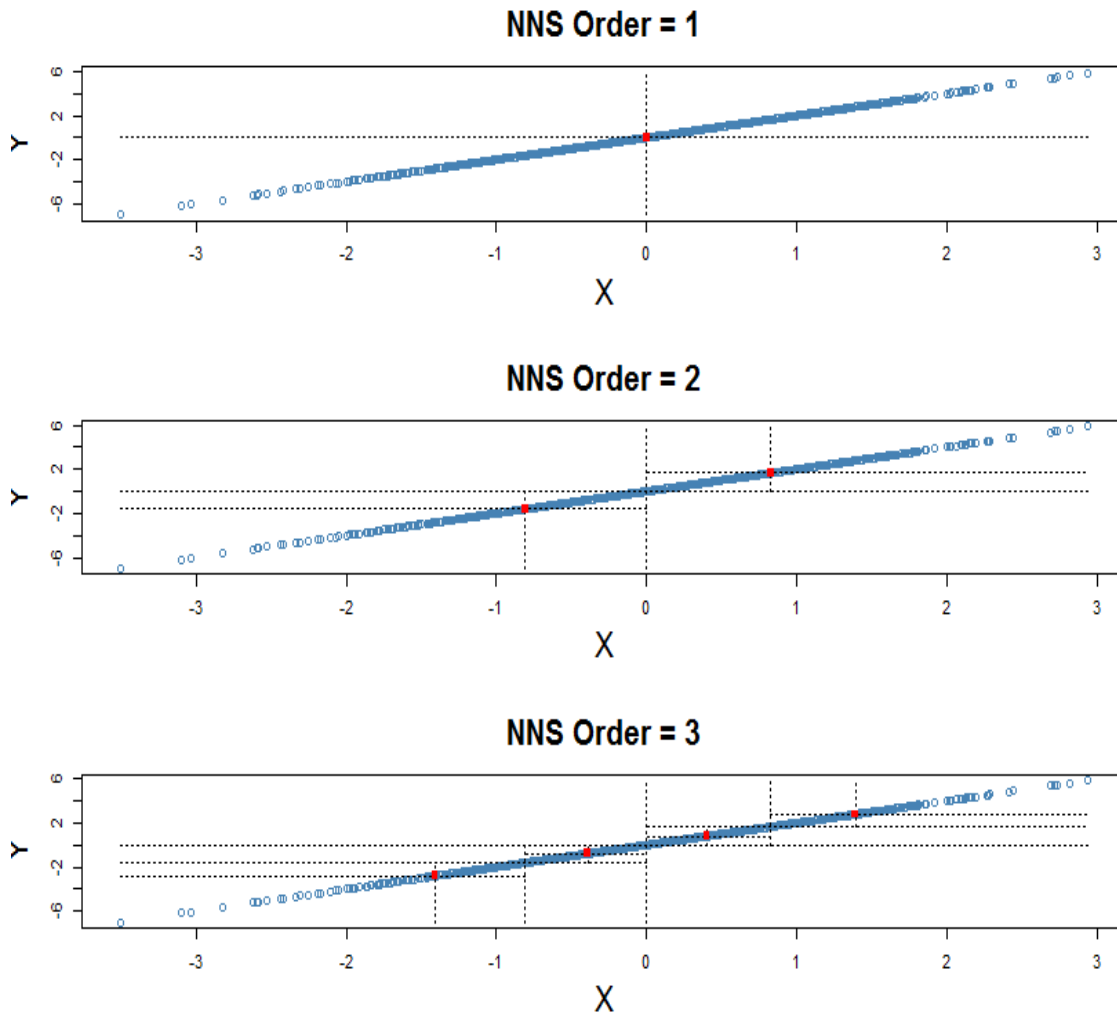


Figure 9. Nonparametric regression points for a linear relationship between (X, Y). Orders progressing restricted to the previous partition boundary.

We can also perform this on nonlinear relationships. Below is an example with 3rd order partitioning, generating a fit to an exponential relationship between the variables.

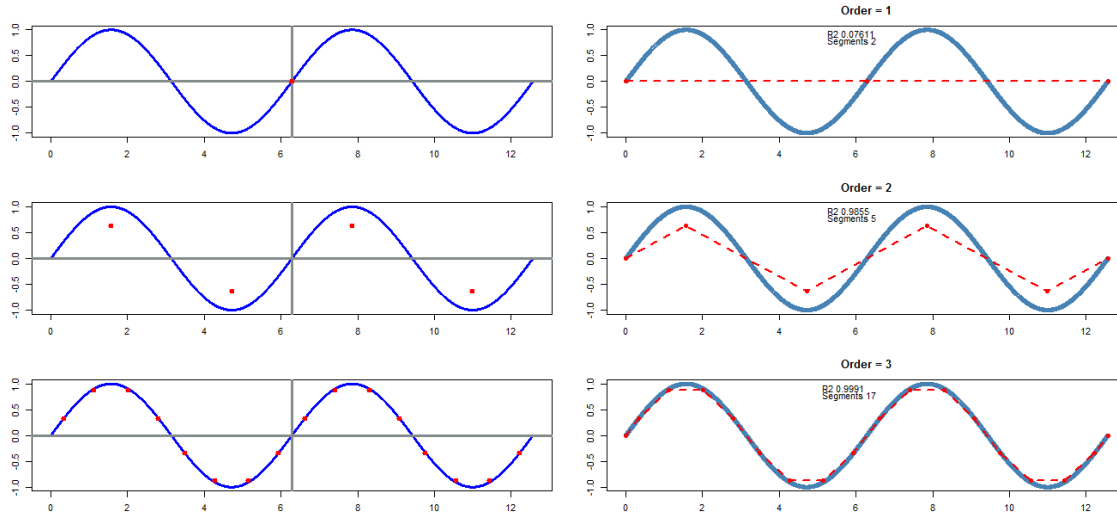


Figure 10. Nonparametric regression points for a nonlinear relationship between (X, Y). As partition orders increase, the curve is better fit.

Generating a multiple variable nonlinear regression analysis requires creating a synthetic variable. This variable, X^* is the weighted average of all of the explanatory variables. The weighting is the nonlinear correlation derived from the $n \times n$ matrix where the explanatory variables are on the same row as the dependent variable which will have a 1.0 self-correlation. Thus, an explanatory variable with zero correlation to the dependent variable will be excluded from consideration.

Thus,

$$X^* = \frac{\sum_{i=1}^n (\rho_{y,x_i})(x_i)}{n} \quad (18)$$

And the nonlinear multiple regression can be performed in kind to the two variable example above with means of Y , X^* as intersection points. This is similar to a nonparametric local means regression only the number of means has to be a factor of 4 due to the four partial moment matrices per each analysis.

Figure 11 below is the nonlinear correlation matrix and the subsequent weightings for the multiple variable nonlinear regression using SPY as the dependent variable with TLT, GLD, FXE, and GSG as explanatory variables.^{iv} The data involved 100 daily observations from 5/8/12 through 9/27/12 for all variables. As shown in Viole and Nawrocki (2012c) partial moments asymptotically converge to the area of the function, and stabilize with approximately 100 observations.

```
> NNS.cor>ReturnsDF,order=3)
```

	GSG	GLD	TLT	FXE	SPY
GSG	1.00000000	-0.10111213	-0.05050505	0.06070809	0.11111111
GLD	-0.10111213	1.00000000	0.23232323	0.21212121	0.03030303
TLT	-0.05050505	0.23232323	1.00000000	0.15151515	-0.23242629
FXE	0.06070809	0.21212121	0.15151515	1.00000000	0.23232323
SPY	0.11111111	0.03030303	-0.23242629	0.23232323	1.00000000

Figure 11. Nonlinear correlation matrix for 5 variables (SPY, TLT, GLD, FXE, GSG). Highlighted row isolates the coefficients for equation 18.

In this example per equation 18 our aggregated explanatory variable is,

$$X^* = \frac{-0.23(TLT) + 0.03(GLD) + 0.23(FXE) + 0.11(GSG)}{4}$$

Again, there are no multicollinearity issues with the explanatory variables, it simply does not matter if they are correlated or not. Below in figure 13 is the graph of this analysis with our 3rd order fit.

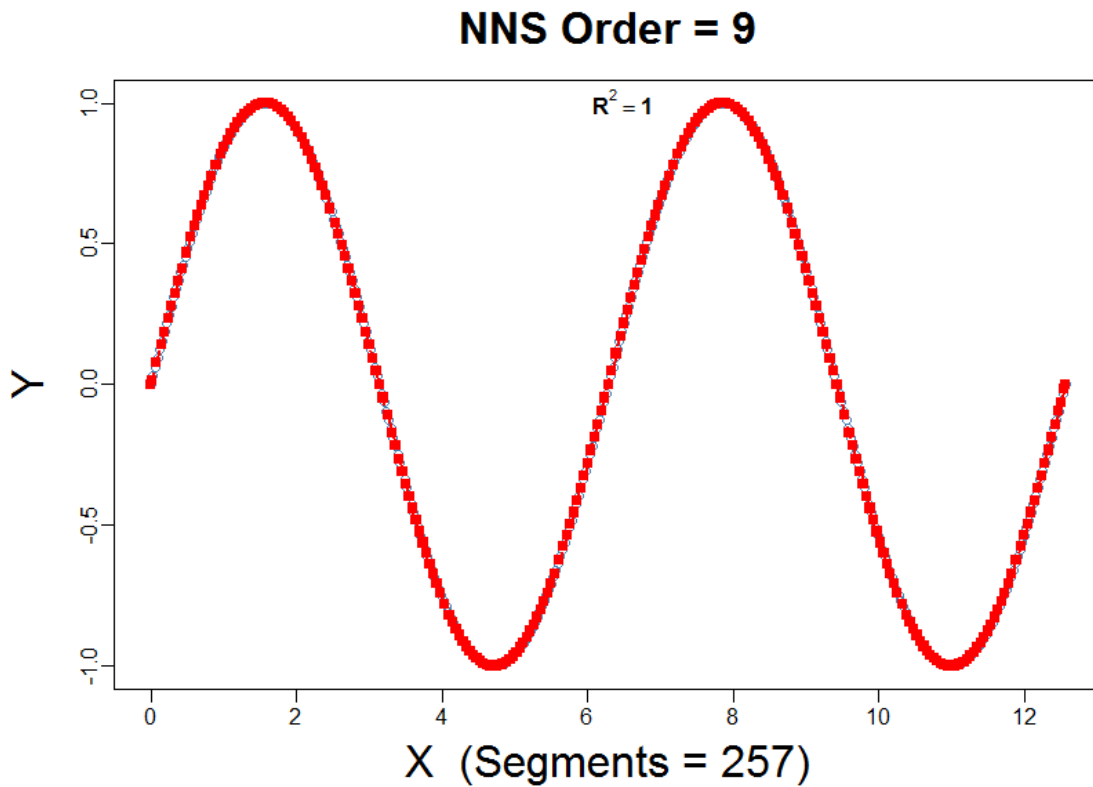


Figure 12. Our 9th order fit for a sine wave function of X.

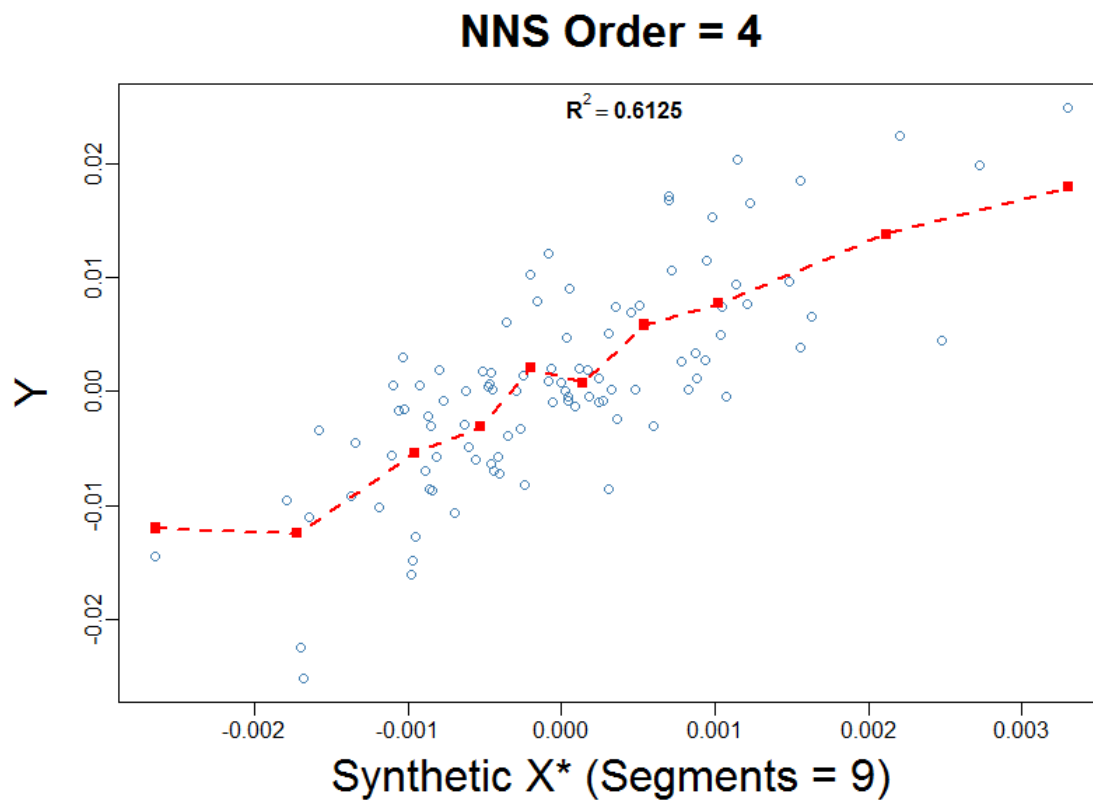


Figure 13. Our 4th order fit for an undetermined function of X^* .

5. DISCUSSION AND SUMMARY:

There is no argument as to why the partition cannot be further specified N times, ultimately yielding a 4^N number of segments. The partial moments are direct computations, just as other statistics such as means and variances. The obvious benefit is the ability to parse what was referred to as “noise” into valid information. Due to the fact that individual observations are weighted by $\left(\frac{1}{T}\right)$, the number of observations in each segment will weigh the segment accordingly; thus affirming outlier observation status for such instances where a segment has minimal occupancy.

The purpose of this paper was to put forth a nonparametric, nonlinear correlation metric where Chen et al. (2010) note, “there is no commonly use statistic quantifying nonlinear correlation that can find a similarly generic use as Pearson’s correlation coefficient for quantifying linear correlation.” Our linear sum of the weighted micro does indeed capture the aggregate correlation. But, unlike Pearson’s single correlation coefficient, we also generate the information necessary to reconstruct the relationship from the individual partial moment matrices. As for a direct policy statement resulting from the nonlinear regression analysis; it would have to assume the form of a conditional equation whereby each linear segment is defined for a specific range of the explanatory variable(s).

References:

Chen, Y. A., Almeida, J., Richards, A., Muller, P., Carroll, R., and Rohrer, B. (2010). "A Nonparametric Approach to Detect Nonlinear Correlation in Gene Expression." *Journal of Computational and Graphical Studies*, Volume 19, Number 3, p. 552-568.

Guthoff, A., Pfingsten, A. and J. Wolf (1997). "On the Compatibility of Value at Risk, Other Risk Concepts, and Expected Utility Maximization"; in: Hipp, C. et.al. (eds.): *Geld, Finanzwirtschaft, Banken und Versicherungen: 1996; Beiträge zum 7. Symposium Geld, Finanzwirtschaft, Banken und Versicherungen an der Universität Karlsruhe vom 11.-13. Dezember 1996*, Karlsruhe 1997, p. 591-614.

Holthausen, D. M. (1981). "A Risk-Return Model With Risk And Return Measured As Deviations From a Target Return." *American Economic Review*, v71(1), 182-188.

Viole, F. and Nawrocki, D. (2012). "Deriving Cumulative Distribution Functions & Probability Density Functions Using Partial Moments." Available at SSRN: <http://ssrn.com/abstract=2148482> .

Endnotes:

ⁱ Newton proved the integral of a point in a continuous distribution to be equal to zero.

ⁱⁱ If no data exists in a subset, no mean is calculated.

ⁱⁱⁱ The horizontal line as in the equation $Y = 1$ (point probability) yields a 0 correlation for both Pearson's correlation and our metric.

^{iv} All variables in the regression are exchange traded funds (ETFs) that trade in US markets: SPY is the S&P 500 ETF, TLT is the Barclays 20+ year Treasury Bond ETF, GLD is the Gold Trust ETF, FXE is the Euro Currency ETF, and GSG is the S&P GSCI Commodity Index ETF.